

A teleportation protocol in Schwarzschild-de Sitter space

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ABSTRACT: We propose a new information transfer protocol for de Sitter space, using black holes as energy reservoirs. We consider antipodal observers in pure de Sitter space in the Bunch-Davis state. They can store Hawking modes from the cosmological horizon in a box. Alternatively, due to thermal fluctuations in de Sitter space, black holes formed through a pair-creation process can be used as energy reservoirs. We focus on the Nariai black hole case, which corresponds to an equilibrium state. Once the black hole is produced, energy pulses can be released into its interior, opening a traversable wormhole. We provide bounds for the amount of information that can be transferred. Specializing in (1+1)-dimensions, we explore how the teleportation protocol leads to an explicit geometric description of the information transmitted through an island region. The protocol uncovers quantum information aspects of de Sitter space, independently of any particular realization of de Sitter space holography.

KEYWORDS: 2D Gravity, Black Holes, de Sitter space

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1 Introduction

In recent years, traversable wormholes (see [1] for a recent review) have provided key insights into how quantum information builds spacetime in the context of the ER=EPR conjecture [2]. To generate a traversable wormhole in any crunching spacetime region, there needs to be a violation of the averaged null energy condition (ANEC) [3]. This condition states that

$$\int_{\Gamma} T_{\mu\nu} k^{\mu} k^{\nu} d\lambda \geq 0, \quad (1.1)$$

where $T_{\mu\nu}$ is the matter stress-energy tensor along complete achronal null geodesics, Γ , which is described by the tangential vectors k^{μ} and the affine parameter λ . This is essentially a statement about causality in general relativity [4–6], which can be violated by quantum matter in curved spacetime [7, 8]. Gao, Jafferis, and Wall (GJW) first proposed a protocol that generates traversable wormholes in BTZ black holes [9]. Several improvements to this protocol have been made in the last few years. Some examples include the stabilization of the wormhole with different matter content [10–14], and applications to probe typical black hole microstates [15, 16]. Although the previous findings rely on the AdS/CFT correspondence, there has been much interest in extending holography to arbitrary types of spacetimes, perhaps even those pertinent for early and late-time cosmology [17–21]. In particular, there is a recent proposal for information exchange in pure dS space [22] which implements a protocol for generating traversable wormholes in de Sitter (dS) space according to a pair of antipodal static patch observers, using the fact that positive energy perturbations elongate the dS

Penrose diagram, as shown by Gao and Wald [23]. The main technical difficulty in adapting the GJW protocol to dS space is it lacks a well-developed holographic dual theory. Thus, one has to rely solely on a bulk perspective, resulting in two major difficulties: assuring the synchronicity of the clocks between the two static observers wanting to exchange information, and controlling the amount of energy required to open up the wormhole.

Purpose of the work. We propose a modification of the GJW protocol for black holes in dS space and derive bounds on the amount of information that can be extracted from the protocol. We first use a protocol described in [22] to allow two causally separated observers to communicate with each other. Although each information exchange between the observers would only allow an amount order $\mathcal{O}(1)$ of bits, and require a dS scrambling time $t_S = \ell \log S_{dS}$ for the message to arrive, the two observers could agree on a binary language akin to Morse code to decode it. In return, this classical communication channel (as in the BTZ case [9]) makes it possible to synchronize their clocks and schedule future events. Secondly, as pointed out in [22], more energy needs to be gathered from the cosmological horizon to increase the amount of information exchange between the two static patches. This can be achieved by placing a one-way transparent, or semi-mirrored (i.e. imposing Dirichlet boundary conditions on one side and transparent boundary conditions on the other one) Dyson sphere (DS) [24] around the center of the static patch $r = 0$, effectively producing an energy reservoir. To guarantee the largest information exchange, we consider a DS of radius $r_{DS} = r_N$, corresponding to the largest black hole radius that can fit into a patch of dS (i.e. a Nariai black hole). After a collecting time $\tilde{T}_N \propto r_N^3$, both observers will agree to let the radiation escape from the DS. On the one hand, the released energy will act as (a series of) positive shock waves emanating from the DS, and result in the formation of a cosmic wormhole [22]. On the other hand, the *lack* of energy bouncing back towards the black hole effectively acts as (a series of) *negative* shocks due to energy conservation, which opens up a wormhole through the black hole region [25].

Alternatively, when waiting long enough, a pair of black holes can spontaneously appear as a consequence of the thermal bath filling empty dS spacetime.¹

As previously mentioned, one of the advantages of our procedure is that it does not require a particular realization of dS space holography.² Moreover, although we explicitly perform the protocol in (3+1)-dimensions to describe how the observers can trap the radiation, a generalization to arbitrary dimensions should be possible, given the generality of the arguments. We will provide bounds on the amount of information that can be exchanged by the observers based on the sizes of the respective wormhole throats.

After obtaining the information bound based on the classical gravitational backreaction, we consider the effects arising from the presence of (entangled) matter fields across the two static patches, leading to the formation of (at least) one island in the black hole region. See [35, 36] for the original formulation of the island rule, and [37] for a pedagogical review. According to this rule, the von Neumann entropy of a weakly gravitating region $\mathbf{R}$ is given

¹See [26] for the original description of the semiclassical instability of dS space, and [27] for the pair creation rate from the no boundary proposal. See also [28–30] for recent discussions about a constrained instanton interpretation of this process.

²We refer the reader to [31–33] for pioneering work in this area, and [34] for a recent review.

by the minimum value among all extrema of the generalized entropy of a region $\mathbf{R} \cup \mathbf{I}$. In other words

$$S(\mathbf{R}) = \min \text{ext}_{\mathbf{I}} S_{\text{gen}}(\mathbf{R} \cup \mathbf{I}), \quad (1.2)$$

where $\mathbf{I}$ is called the island region found by extremizing the functional

$$S_{\text{gen}}(\mathbf{R} \cup \mathbf{I}) = \frac{A(\partial \mathbf{I})}{4G_N} + S_{\text{matter}}(\mathbf{R} \cup \mathbf{I}). \quad (1.3)$$

In this equation, $S_{\text{matter}}(\mathbf{R} \cup \mathbf{I})$ denotes the von Neumann entropy of the bulk matter fields, and $A(\partial \mathbf{I})$ is the codimension-2 area functional evaluated at the endpoints of the region $\mathbf{I}$.

The observer on the region $\mathbf{R}$ can perform a series of decoding channels from the Hawking radiation, such as the Petz map [38–43], reconstruct the entanglement wedge that includes the island region. In this case, information is sent to the island region and later collected by low-energy observers.³ Although the island rule is less developed in the context of dS space, there are indications that islands, seen as disconnected entanglement wedges, will appear in general types of backgrounds [46, 47].

To make an explicit realization of the above protocol, we employ the s-wave dimensional reduction of a near-Nariai black hole background, called the full-reduction dS Jackiw-Teitelboim gravity (JT) gravity model. See [17, 18, 48–59] for different developments on the island rule in dS JT gravity. The antipodal static patch observers will prepare the shockwaves from the thermal bath. We then locate a pair of detectors at $\mathcal{I}^+$ of the inflating regions, representing meta observers, to collect Hawking modes coming from the black hole, as shown in figure 5. In this situation, the effect of the shockwave is to shift the location of the island with respect to the one without the backreaction. The antipodal observers can transmit information about the island region directly to the detectors (i.e. the metaobservers).

Our *goal* is then to derive bounds on the information transfer in near-Nariai black hole backgrounds, independent of dS holographic assumptions, using the GJW protocol, as well as from the island rule in the s-wave reduction of the near-Nariai geometry with CFT₂ bulk matter fields.

Outline. The structure of the manuscript is as follows. In section 2, we review some of the basics on state preparation and SdS black holes to carry out our protocol. We will then focus mostly on (3+1)-spacetime dimensions to make concrete statements on how to trap the radiation. We find bounds on the amount of information that can be transferred between static patch observers. Section 3 is devoted to the (1+1)-dimensional formulation of the protocol from dS JT gravity where we add metaobservers in the inflationary patch of the near-Nariai spacetime equipped with detectors. We show that the traversable wormhole protocol can probe the island region according to the metaobservers. In section 4, we include a summary of our proposal and we conclude with some future directions. For the convenience of the reader, appendix A reviews the Hayden-Preskill (HP) protocol to perform information recovery, which we apply in our protocol. Meanwhile, appendix B contains a review of the

³Related problems on information transmission through wormholes and islands have been addressed in the DS/dS context [44, 45].

s-wave reduction of SdS black holes in higher dimensions to generate the full-reduction dS JT gravity theory. Lastly, appendix C contains a discussion about the small shift approximation in the shockwave analysis in the weakly coupled gravity regime.

2 General protocol

2.1 State preparation

Our starting configuration follows the considerations in [22, 60, 61] for constructing the Bunch-Davis state of pure dS space. We begin studying a single static patch region. Take for example the south wedge, S . In the perturbative quantum gravity regime, we can define a Hilbert space with energy eigenstates, $|E_i\rangle_S \in \mathcal{H}_S$ with respect to a one-sided modular Hamiltonian H_S . One can also define a natural thermal vacuum state, known as the Bunch-Davis state by introducing a decoupled antipodal static patch region. Let us denote it by the north, N , with energy eigenstates $|E_i\rangle_N \in \mathcal{H}_N$ for the Hamiltonian H_N . We may generate the cyclic, separating vacuum of the theory by means of the thermofield double construction of the maximally entangled state with respect to S and N [60],

$$|\Psi_{\text{BD}}\rangle = \frac{1}{\sqrt{Z}} \sum_i \exp\left(-\frac{1}{2}\beta E_i\right) |E_i\rangle_S \otimes |E_i\rangle_N^* \in \mathcal{H}_S \times \mathcal{H}_N. \quad (2.1)$$

The symbol $*$ denotes complex conjugation, and β refers to the inverse temperature of both patches. The Bunch-Davis state is the vacuum of the time automorphism generator, $H_{\text{grav}} = H_N - H_S$, where the minus sign indicates the difference in time orientations between antipodal patches; and importantly, $H_{\text{grav}} |\Psi_{\text{BD}}\rangle = 0$. This construction has been recently discussed in the context of the algebra of observables for dS space in [60] in the presence of matter and weakly coupled gravity (see [62–68] for related developments). The initial configuration is illustrated in figure 1. The pair of antipodal observers are then prepared in a maximally entangled state and they possess a classical communication channel, so we will assume that their internal clocks are synchronized as we start the protocol.

2.2 Black holes in dS space

The main motivation in this work is to incorporate the appearance of black holes into an information exchange protocol in dS space. We will allow the observers to construct an energy reservoir from the Hawking radiation coming out of the cosmological horizon in figure 1. The reservoir can then collapse into an SdS black hole connecting the antipodal observers. We will specify the exact conditions for this to happen in section 2.4, after introducing some general notions in this section. Alternatively, we may also consider the semiclassical instability process of dS space that leads to pair production of SdS black holes [26].

SdS black holes in $(d+1)$ -dimensions are spherically symmetric vacuum solutions to Einstein equations with a positive cosmological constant $\Lambda > 0$. These configurations can be described with the following metric in static patch coordinates

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega_{d-1}^2, \quad \text{with} \quad (2.2)$$

$$f(r) = 1 - \frac{r^2}{\ell^2} - \frac{16\pi G_N M}{(d-1)\Omega_{d-1} r^{d-2}}. \quad (2.3)$$

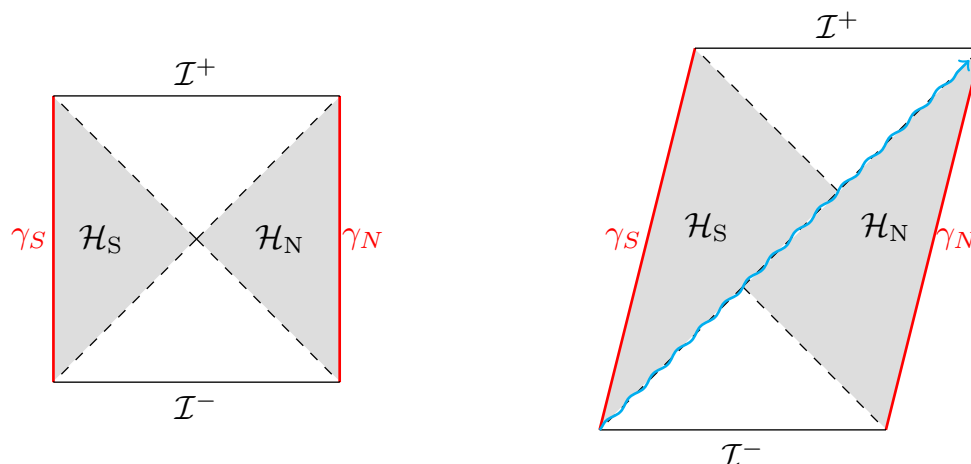


Figure 1. (a) dS space as seen by a pair of antipodal observers following a worldline γ_N and γ_S (in red) for the Bunch-Davis preparation of state described as a thermofield double state, with respective gravitational Hilbert spaces $\mathcal{H}_{N/S}$. (b) The spacetime gets elongated after the insertion of a positive energy shockwave (cyan wavy line) [23]. The dashed black line indicates the location of the cosmological horizon.

Here, M is the black hole mass parameter, $\ell^2 = d(d-1)/(2\Lambda)$ is the dS curvature radius, G_N is Newton's constant and $\Omega_{d-1} = 2\pi^{d/2}/\Gamma(d/2)$ is the volume of a unit $(d-1)$ -sphere. In (2.2), we have performed a gauge fixing choice where the static patch time of the N and S pole observers are described as $t = t_S = t_N$. On the other hand, the parameter M cannot be arbitrarily large, as this can result in complex roots for the blackening factor $f(r)$, which would correspond to a naked singularity. Such structures are forbidden in nature according to the cosmic censorship conjecture; see [69] for a review. Therefore, we consider a range for the mass of SdS black holes $M \in [0, M_N]$. For $M = 0$, one recovers empty dS space, with a cosmological horizon located at $r = \ell$. The maximal mass $M_N = \frac{d-1}{d} \frac{\Omega_{d-1}}{8\pi G_N} r_N^{d-2}$ is referred to as the Nariai mass. This limit corresponds to the extreme case where the radii of the black hole and cosmological horizon coincide, $r_b = r_c = r_N = \ell \sqrt{\frac{d-2}{d}}$, which is called the Nariai radius [26, 70].⁴ For our purposes, we describe the temperature, T , and the thermodynamic entropy, S , of the cosmological, c, and black hole, b, horizons by

$$T_{b,c} = \frac{\kappa_{b,c}}{2\pi}, \quad S_{b,c} = \frac{\mathcal{A}_{b,c}}{4G_N}, \quad (2.4)$$

where $\kappa_{b,c}$ and $\mathcal{A}_{b,c} = \Omega_{d-1} r_{b,c}^{d-1}$ denote the surface gravities and areas of b and c respectively. For the surface gravity, we employ the same normalizations as in [27] which guarantee that the black hole and cosmological horizons are in thermal equilibrium for the Nariai configuration, where the inverse temperature (2.4) in the Nariai limit is $\beta_N = \frac{2\pi\ell}{\sqrt{d}}$.

2.3 Preparation of the black hole

We now discuss the conditions to produce the near-Nariai black hole by collecting radiation in a thermal bath in dS space. We model the bath as a Dyson sphere (i.e. a boundary

⁴The reader might check [29] for further details on the quasilocal thermodynamics of these solutions.

wall that absorbs radiation) available to each of the static patch observers collecting the radiation coming from the cosmological horizon before sending the message. When trapping all the radiation into a box (i.e. the Dyson sphere) of size $r_{\text{box}} \leq r_N$, one can compute the time it would take to build a reservoir of any energy $E_{\text{res}} \leq M_N$ using Stefan-Boltzmann law in $(3 + 1)$ -dimensions,

$$\mathcal{P}_c = \sigma \mathcal{A}_c T_c^4 \quad (2.5)$$

with $\mathcal{P}_c = dM/dt$ the radiated power, σ the Stefan-Boltzmann constant, and $\mathcal{A}_c$ the surface area of the cosmological horizon. Inserting the cosmological temperature, as well as the mass written in terms of the horizon radius r_c

$$T_c = \frac{r_c^2/r_N^2 - 1}{4\pi r_c \sqrt{1 - (M/M_N)^{2/3}}}, \quad M = \frac{r_c}{2G_N} \left(1 - \left(\frac{r_c}{\ell} \right)^2 \right), \quad (2.6)$$

allows us to find

$$\frac{dr_c}{dt} = \frac{G_N \sigma (\ell^2 - 3r_c^2)^3}{8\pi^3 \left(2\ell^2 r_c - 3\sqrt[3]{2} r_c^{5/3} (\ell^2 - r_c^2)^{2/3} \right)^2} \equiv \dot{r}_c. \quad (2.7)$$

The time, $\tilde{\mathcal{T}}_N$, necessary to fill up the whole dS space (till its radius matches the Nariai one) is therefore given by

$$\tilde{\mathcal{T}}_N = \int_{\ell}^{r_N} \frac{dr_c}{\dot{r}_c} \approx \frac{30\sqrt{\pi}}{G_N} r_N^3, \quad (2.8)$$

where we inserted $\sigma = \frac{2\pi^5}{15}$. The evolution of the cosmological and possible⁵ black hole radii in this process are illustrated in figure 3(a).

2.4 Shockwave preparation

To produce a traversable wormhole in a near-Nariai black hole spacetime, the distance between the observer and the horizon has to *grow*. For the inflating patch, this implies that the cosmological horizon radius has to increase, which is only achieved by an excess of positive energy leaking out of the considered patch or equivalently increasing its negative energy density. This implies that we send negative energy in the opposite direction, into the black hole's interior. See figure 2 for an illustration of this step in the protocol.⁶ The deficit of positive or negative energy particles from one patch implies that there would be an excess of opposite sign energy in the opposite direction.

We can treat this step of the protocol as in the previous section, but now the black hole and cosmological fluxes are not restricted by any reflective boundary or Dyson sphere. For a general SdS black hole, the total power passing through the cosmological horizon equals

$$\mathcal{P}_{\text{tot}} = \mathcal{P}_c - \mathcal{P}_b. \quad (2.9)$$

⁵In order to form a black hole, the energy density must be sufficiently large. It would require careful tuning of the position of the reflective boundary. However, we avoid any of these difficulties by filling up a box of size r_N , guaranteeing that we have a black hole by the end of the process.

⁶Throughout the text, we employ the convention for the Penrose diagrams with shockwave shifts in which light signals are discontinuous along a path with the (cosmological or black hole) horizons remaining continuous.

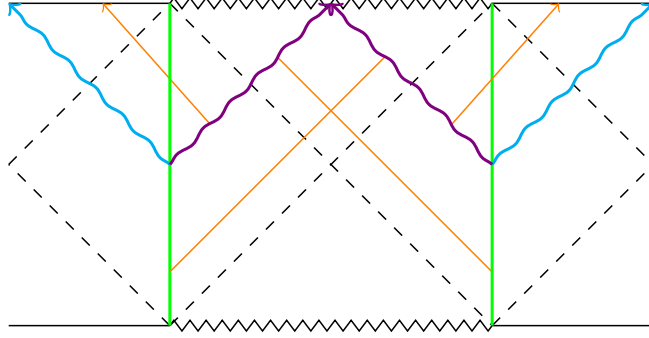


Figure 2. Configuration where the antipodal observers (in green) produce a thermal bath from Hawking radiation that they initially stored in the empty dS preparation of state. Each observer releases a positive energy pulse (cyan) to the inflating region, which also results in a negative energy pulse to the black hole interior (purple). This allows the observers to transmit light signals (orange) to each other.

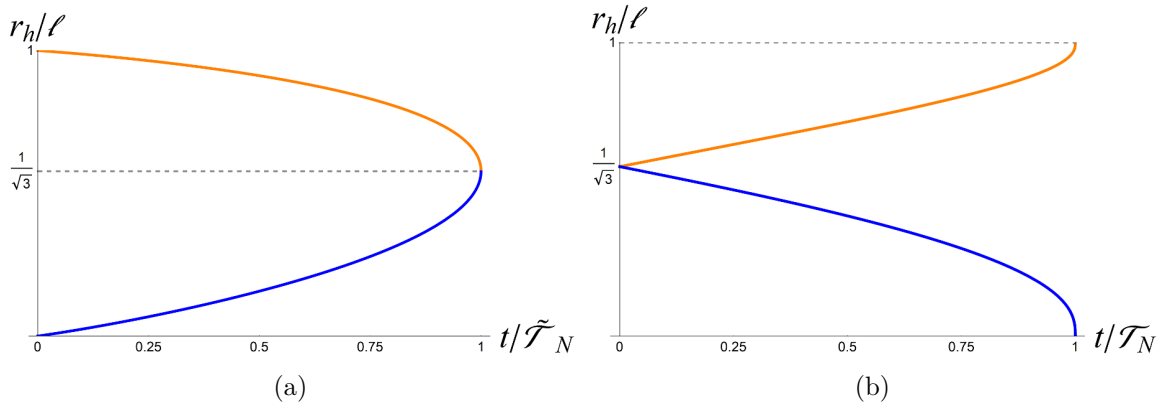


Figure 3. The horizon radii r_h as a function of the static patch time in $d = 3$. The cosmological radius r_c is shown in orange, and the black hole horizon radius is in blue. (a) We start from empty dS space and consider absorbing boundary conditions for trapping the radiation. The relation $\ell^2 = r_b^2 + r_c^2 + r_b r_c$ allows us to plot the black hole radius that could be formed with the gathered energy. (b) We start with a Nariai black hole configuration and let the radiation freely escape through the cosmological horizon.

The lifetime of a four-dimensional Nariai black hole, $\mathcal{T}_N$, can be computed in a similar way as section 2.3, resulting in

$$\mathcal{T}_N \approx \frac{\tilde{\mathcal{T}}_N}{\sqrt{2}}. \quad (2.10)$$

The respective evolution is shown in figure 3(b).

In the strict Nariai limit $\mathcal{P}_{\text{tot}} = 0$. However, when the black hole mass is very close to the Nariai mass $M = M_N(1 - \epsilon_N)$, where $0 < \epsilon_N \ll 1$, one can find that at first order

$$\mathcal{P}_{\text{tot}}(M \rightarrow M_N) = \mathcal{P}_{\text{tot}}(M_N) + \epsilon_N \left(\frac{\partial \mathcal{P}_{\text{tot}}}{\partial \epsilon_N} \right)_{\epsilon_N=0} = \frac{\epsilon_N}{3\pi^3 \ell^2}. \quad (2.11)$$

Then, the energy E_N released when opening the box for a duration $\delta\tau$ for a SdS black hole close to the Nariai limit is given by

$$E_N = \frac{\epsilon_N}{3\pi^3\ell} \left(\frac{\delta\tau}{\ell} \right). \quad (2.12)$$

Such emission of energy can cause a run-away effect leading to the complete evaporation of the black hole. Notice that imposing $\delta\tau/\ell \equiv \epsilon_\tau \ll 1$, the released energy ratio E/M by the pulse in the Nariai limit can be approximated by

$$\frac{E_N}{M_N} \approx \epsilon_N \epsilon_\tau \left(\frac{\ell_{\text{pl}}}{\ell} \right)^2. \quad (2.13)$$

The result is an extremely small quantity, therefore justifying a shockwave description.

2.5 GJW protocol for Nariai spacetimes

As the observers throw matter, they are able to generate a traversable wormhole in the black hole interior, as well as in the inflating region. In that way, the observers can transmit light signals through the wormhole, as shown in figure 2.⁷ Notice that we do not require a holographic theory to establish this step in the protocol.⁸

To make the previous statements more precise, we calculate below the bounds on the amount of information transmitted during the process.

- For the *black hole patch*, we focus on the static patch time that the message spends behind the horizon of figure 2, given by

$$\Delta t = t_{\text{lc}} + t_{\text{shift}} + \delta t, \quad (2.14)$$

where t_{lc} stands for the light-crossing time, t_{shift} corresponds to the time for gathering the required energy and δt is the time delay between collecting the energy and sending the message.

For the gathering time, t_{shift} , we consider the reflection time for a photon to go from the radius of the Nariai black hole ($r = r_N$) to the location of the static patch observer at $r = 0$, which is given by:

$$t_{\text{shift}} \approx \ell \log S_b, \quad (2.15)$$

where $S_b = S_N/2$ is the Nariai black hole entropy. In the following, we consider the minimum possible time for information recovery, where t_{lc} and δt can be neglected. This implies that the time difference before and after the absorption of photons, where the black hole entropy increases due to the presence of the shockwave as $S_b \rightarrow S_b + S_{\text{Shock}}$, is given by

$$\Delta t \propto \log[S_b + S_{\text{Shock}}]. \quad (2.16)$$

⁷Operators supported in previously causally disconnected spacetime points in the static patches of the black hole and cosmological regions become causally connected. This modifies the commutation relations of the algebra of observables [71]. We thank Aron Wall for important comments about this point.

⁸A key difference with respect to realizing the protocol in an asymptotically AdS black hole spacetime is that we do not need to violate ANEC in order to open the wormhole, as we have elaborated. We thank Ben Freivogel for his useful question about this point.

In that case, the recovery time is given as

$$\Delta\tau \equiv \Delta t - \tau_b^* \propto \log \left[1 + \frac{S_{\text{Shock}}}{S_b} \right], \quad (2.17)$$

where $\tau_b^* = \ell \log S_b$ is the black hole scrambling time.

The condition for the wormhole to be traversable requires the recovery time to be smaller than the original scrambling time ($\Delta\tau < 0$), such that

$$-S_b < S_{\text{Shock}} < 0. \quad (2.18)$$

The result agrees with the fact that the shock wave should have the opposite sign than the energy of the black hole, as this kind of shock-wave *reduces* its entropy.

- For the *inflating patch*, we may use the same expression as before, $\Delta t = t_{\text{lc}} + t_{\text{shift}} + \delta t$, such that, neglecting the contribution of δt , the recovery time is now given as follows

$$\Delta t = t_{\text{lc}} + t_{\text{shift}} \propto \log[S_c - S_{\text{Shock}}]. \quad (2.19)$$

Therefore, the difference in emission times is then

$$\Delta\tau = \Delta t - \tau_c^* \propto \log \left[1 - \frac{S_{\text{Shock}}}{S_c} \right]. \quad (2.20)$$

with $\tau_c^* = \ell \log S_c$. We again notice that $\Delta\tau < 0$ is necessary for a traversable wormhole, which requires

$$0 < S_{\text{Shock}} < S_c. \quad (2.21)$$

This result reflects the fact that the shockwave has to increase the entropy of the cosmological horizon to generate the traversable wormhole. It also corroborates that the Gibbons-Hawking entropy is counting the degrees of freedom behind the horizon.

As argued in [3, 25], regardless of the protocol, the maximal amount of information that can be transmitted through the wormhole in this context is bounded by

$$N_{\text{bits}} \lesssim r_N^{d-1}/G_N \approx S_b, \quad (2.22)$$

given that we cannot transfer more bits of information than those available from the surface area of the black hole horizon, as it gets destroyed during the process. See appendix A for an explanation of the relation between the number of bits and the criterion for successful information recovery. This suggests that the amount of information passing through the resulting wormhole throats in SdS space can be estimated by considering the differences

$$\Delta S_b = S_b(M) - S_b(M - E) \quad (2.23)$$

$$\Delta S_c = S_c(M - E) - S_c(M), \quad (2.24)$$

where $E = \int \mathcal{P}_{\text{tot}} dt$ is the total energy exchanged during the protocol. Notice that the maximal total information that can be transmitted in this way is given by

$$\Delta S_{\text{tot}} = \Delta S_b + \Delta S_c = S_N/2 + (S_{\text{dS}} - S_N/2) = S_{\text{dS}}, \quad (2.25)$$

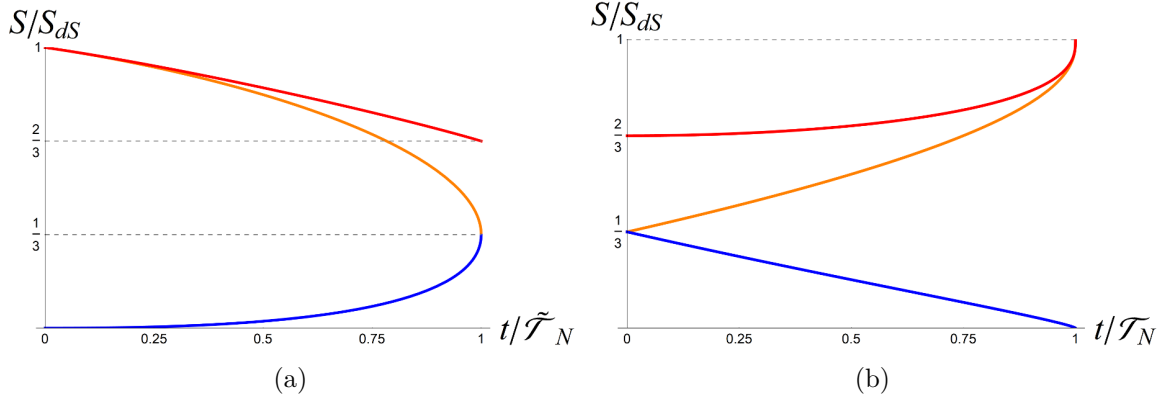


Figure 4. The cosmological entropy S_c (orange), black hole entropy S_b (blue), and their sum, the total entropy (red), as a function of time in $d = 3$. (a) Empty dS with absorbing boundary conditions from the Dyson sphere to capture the energy. (b) Nariai spacetime where the energy flux freely escapes through the cosmological horizon.

with $E = M_N$; whereas the total entropy difference is $\Delta S = S_{dS} - S_N = S_N/2$ in (3+1)-dimensions. Furthermore, assuming small energy ratios $E/M \equiv \epsilon$ and expanding at first order, we find that

$$s_b \equiv \epsilon \left(\frac{\partial S_b}{\partial \epsilon} \right)_{\epsilon=0} = \epsilon M \frac{\partial S_b}{\partial M} = E \beta_b, \quad (2.26)$$

$$s_c \equiv -\epsilon \left(\frac{\partial S_c}{\partial \epsilon} \right)_{\epsilon=0} = -\epsilon M \frac{\partial S_c}{\partial M} = E \beta_c, \quad (2.27)$$

where we normalized the inverse temperature β with respect to the observers at $r_{\mathcal{O}}$. Notice that the wormhole sizes do not coincide, i.e. $s_b \neq s_c$, for any M except the Nariai mass M_N , where $\beta_b = \beta_c = 2\pi r_N$. Figure 4 shows the evolution of the entropies as a function of time.

3 Information recovery in (1+1)-dimensions

In this section, we incorporate semiclassical effects into our protocol. For that, we adopt dS JT gravity from the S-wave reduction of the near-Nariai black holes.⁹ The theory is defined by the action

$$I_{\text{dS JT}} = \frac{\phi_0}{16\pi G_2} \int_{\mathcal{M}} d^2x \sqrt{-g} R + \frac{1}{16\pi G_2} \int_{\mathcal{M}} d^2x \sqrt{-g} \left(R - \frac{2}{\ell^2} \right) \phi + \frac{1}{8\pi G_2} \int_{\partial\mathcal{M}} dy \sqrt{-h} \phi K + I_{\text{matter}}, \quad (3.1)$$

where ϕ is the dilaton field, which describes scalar mode perturbations around the Nariai black hole solution, G_2 the effective gravitational coupling, and I_{matter} denotes the bulk CFT₂ theory. See appendix B for more details on deriving this theory from the dimensional reduction

⁹The island rule has been derived in [72] for the type of model under consideration, albeit for static backgrounds. However, our findings point out that the island still obeys maximin conditions [73], and the shockwave simply shifts its location.

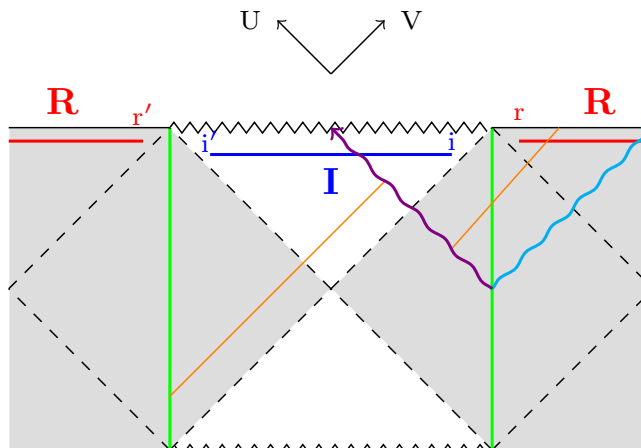


Figure 5. Proposal for the teleportation protocol in the dS JT gravity. Same color labeling as in figure 2. The signals sent through the wormhole arrive at a pair of detectors, $\mathbf{R}$, in an interval bounded by the points $[r, r']$ near $\mathcal{I}^+$ of the inflating patch. The radiation is purified by the island, $\mathbf{I}$ (blue), which is bounded by the points $[i, i']$. The axes for the Kruskal coordinates (U, V) are also shown. The gray area represents the region covered by a single Kruskal coordinate system, which includes the static patches and either the black hole or the inflating regions.

of a near-Nariai black hole. The equations of motion, corresponding to (3.1), are given by

$$-\nabla_\mu \nabla_\nu \phi + g_{\mu\nu} \square \phi + \frac{\phi}{\ell^2} g_{\mu\nu} = 8\pi G_2 \langle T_{\mu\nu} \rangle, \quad (3.2)$$

with $\langle T_{\mu\nu} \rangle$ the expectation value of the stress tensor.

Islands before the shockwave. We now place detectors near the $\mathcal{I}^+$ region of the inflationary patches. The metaobservers can then collect Hawking modes in weakly gravitating regions. See figure 5 for an illustration of the proposal. In the configuration of interest, we evaluate the generalized entropy (1.3) considering that the state on a complete Cauchy slice is pure at a given time,¹⁰ the state on $\mathbf{R} \cup \mathbf{I}$ is purified by that on the complement region, resulting in $S_{\text{gen}}(\mathbf{R} \cup \mathbf{I}) = S_{\text{gen}}(\overline{\mathbf{R} \cup \mathbf{I}})$. We consider a completely symmetric configuration for the endpoints of the island and those of the detectors in $\mathbf{R}$. Let us introduce conformal vacuum coordinates $x^\pm$. The metric has the form

$$ds^2 = -\Omega(x^+, x^-)^{-2} dx^+ dx^-, \quad (3.3)$$

where we denote $\Omega(x^+, x^-)$ as the conformal factor.

The von Neumann entropy for a single interval $[j, k]$ is given by

$$S_{\text{vN}}([j, k]) = \frac{c}{6} \log \left[\frac{(x_j^+ - x_k^+)(x_j^- - x_k^-)}{\Omega(x_j^+, x_j^-)\Omega(x_k^+, x_k^-)\epsilon_j\epsilon_k} \right], \quad (3.4)$$

¹⁰Notice that in our protocol, only the global state of the bulk geometry together with the energy pulses is pure. This occurs since we prepared the Bunch-Davis state in pure dS space, and later trapped some of the matter fields into a reservoir carried by the antipodal observers.

where the factors ϵ_j and ϵ_k denote the usual short-distance cutoffs of the von Neumann entropy. Meanwhile, the dilaton $\phi(\partial I)$ gives the area contribution in (1.3),

$$S_{\text{gen}}(\mathbf{R} \cup \mathbf{I}) = \frac{\phi(\partial \mathbf{I})}{4G_N} + S_{\text{matter}}(\mathbf{R} \cup \mathbf{I}) . \quad (3.5)$$

We use Kruskal coordinates (V, U) to denote the metric and the *sourceless* dilaton

$$\begin{aligned} ds^2 &= -\frac{4\ell^4}{(\ell^2 - UV)^2} dU dV , \\ \phi(U, V) &= \phi_r \frac{\ell^2 + UV}{\ell^2 - UV} . \end{aligned} \quad (3.6)$$

Let (U_i, V_i) denote one of the endpoints of $\mathbf{I}$, and (U_r, V_r) one of the endpoint in $\mathbf{R}$, as shown in figure 5. Islands in the Bunch-Davis vacuum dS JT gravity have been derived in many places throughout of the literature in the limit that the spatial interval distance of regions $\mathbf{R}$ and $\mathbf{I}$ is large compared to the distance between separating them (see [17, 72] for example). We need an analytic continuation in the Kruskal coordinate system to evaluate the von Neumann entropy given that $\mathbf{I}$ and $\mathbf{R}$ are located in the black hole and inflating patches. The metric (3.6) in global conformal coordinates is given by

$$ds^2 = \ell^2 \sec^2 \sigma (-d\sigma^2 + d\varphi^2) . \quad (3.7)$$

Now, one needs to make a shift of coordinates $\varphi \rightarrow \varphi + \pi$ to connect points in the adjacent hyperbolic patches. This amounts to shifting the point (V_r, U_r) as

$$V_r \rightarrow -\frac{\ell^2}{V_r}, \quad U_r \rightarrow -\frac{\ell^2}{U_r} . \quad (3.8)$$

For the region below the shockwave in figure 5, the generalized entropy can be then found as

$$S_{\text{gen}} = \frac{1}{2G_2} \left(\phi_0 - \phi_r \frac{\ell^2 + U_i V_i}{\ell^2 - U_i V_i} + \frac{cG_N}{3} \right) + \frac{c}{3} \log \frac{4\ell^2(\ell^2 + U_i U_r)(\ell^2 + V_i V_r)}{\epsilon_i \epsilon_r (\ell^2 - U_i V_i)(\ell^2 - U_r V_r)} . \quad (3.9)$$

The only island space-like separated from (V_r, U_r) is given by

$$V_i = \frac{U_r \epsilon}{3} + \mathcal{O}(\epsilon^2), \quad U_i = \frac{V_r \epsilon}{3} + \mathcal{O}(\epsilon^2), \quad (3.10)$$

where $\epsilon \equiv \frac{G_2 c}{\phi_r}$ is a small parameter in the semiclassical regime, i.e. $\epsilon \ll 1$.

However, in the extremization of the generalized entropy (1.3) one must also include the possibility for the island to be the empty set, $\emptyset$. In that case, we can produce a Page curve transition between the trivial and non-trivial island (3.10) saddles. One can derive that this transition occurs in global coordinates (3.7) for a detector in the interval $[r, r']$ for $\varphi_r = -\varphi_{r'} \equiv \varphi_{\text{Page}}$ and $\sigma_R \simeq \frac{\pi}{2}$ constant in figure 5, and it is given by [17]

$$\tan \varphi_{\text{Page}} \propto \exp \left[\frac{3}{2G_2 c} \left(\phi_0 + \phi_r + \frac{cG_2}{3} \right) \right] . \quad (3.11)$$

Adding the shockwaves. Now, we allow the observers to release negative energy pulses toward the black hole's interior. Consider for instance the south pole observer in figure 5. The stress tensor is given by¹¹

$$\langle T_{VV}(V) \rangle = \alpha \frac{\delta(V - V_S)}{V_S}, \quad (3.13)$$

which describes a shockwave moving towards the black hole interior along the null ray $V = V_S$. As in section 2, we employ $\alpha < 0$ to form a traversable wormhole on the black hole interior, and we consider that $|\alpha| \ll 1$. See further comments on the small shift approximation in appendix C.

Next, we would like to compare the island location with the shockwave insertion in the protocol, to inquire under what conditions the traversable wormhole enters in causal contact with the island region. Given the stress tensor (3.13), the dilaton solution in (3.6) would be modified according to the graviton equation of motion in (3.1), leading to

$$\phi = \phi_0 - \phi_r \frac{\ell^2 + U_i V_i}{\ell^2 - U_i V_i} + \frac{cG_2}{3} - \frac{8\pi G_2 \alpha}{V_S} \frac{\ell^2 - V_S U_i}{\ell^2 - V_i U_i} (V_i - V_S) \theta(V_i - V_S). \quad (3.14)$$

We want to identify null congruences, which in JT gravity are described by curves satisfying $\partial_{\pm} \Phi = 0$ [22]. We denote them as follows

$$\gamma: \quad V = V_S \frac{2G_2 \alpha}{\phi_r} + \mathcal{O}(\alpha^3), \quad (3.15)$$

$$\delta^+: \quad U = \frac{\ell^2}{V_S} \frac{\phi_r}{2G_2 \alpha} + \mathcal{O}(\alpha), \quad (3.16)$$

$$\delta^-: \quad U = -\frac{\ell^2}{V_S} \frac{2G_2 \alpha}{\phi_r} + \mathcal{O}(\alpha^3). \quad (3.17)$$

The apparent horizon is the outermost surface, with $\partial_+ \Phi = 0$ and $\partial_- \Phi < 0$ [52], which is allowed by both of the curves $\delta^{\pm}$. The curve describing the null shift for $\alpha < 0$, which we implement in the protocol to violate ANEC, corresponds to the one that reduces the size of the black hole, which is δ^- for $V > V_S$. This means that the null shift of the apparent horizon to leading order in α is given by (3.17).

In our conventions, the horizon in the Penrose diagrams remains fixed while the light signals are shifted. As the observer in the north pole sends a light signal to the causal diamond of the island **I** in figure 5, it will get shifted by (3.17), and it will reach the location **R** of the adjacent hyperbolic slice. The signal sent along $U = U_{\text{ls}}$ enters the inflating patch along the line:

$$U = U_{\alpha} \equiv \frac{\ell^2}{U_{\text{ls}}} + \frac{8\pi G_2 |\alpha|}{4\pi \phi_r}, \quad (3.18)$$

¹¹It is possible to set a more physical situation in which the stress tensor becomes a smooth function. This can be done by adding a smearing function $f(V)$, and consider

$$\langle \tilde{T}_{VV} \rangle = \int dV \langle T_{VV} \rangle f(V). \quad (3.12)$$

In general, we expect that the specific form of smearing function depends on the details of the theory. The resulting effect will cause the stress tensor not to change the geometry abruptly [74].

which follows from the analytic continuation in (3.8). Then, the light ray would intercept the causal diamond of the island, found in (3.10), as long as $U_r < U_\alpha$.

Repeating the protocol. As we noticed in (2.13), the energy released as shockwaves in our protocol is generically small, as it is suppressed by a ℓ_{pl}^2 factor. The static patch observers can then decide on performing multiple shockwave protocols, as long as it obeys the bound found in (2.25). In this situation, one must account for backreaction on the island location. Given the presence of the stress tensor in (3.13), one must find the map to vacuum coordinates $x^\pm$ in (3.4). We can find such a map using the conformal anomaly of the stress tensor

$$\langle T_{VV}(V) \rangle = -\frac{c}{24\pi} \{x^+, V\}. \quad (3.19)$$

Taking the Ansatz

$$x^+ = V + \alpha f(V), \quad x^- = U, \quad (3.20)$$

for $\alpha \ll 1$ we obtain the following relation at leading order in α

$$-\frac{c}{24\pi} f'''(V) = \frac{1}{V_S} \delta(V - V_S), \quad (3.21)$$

where the solution to the above differential equation is found to be

$$f(V) = -\frac{12\pi}{c V_S} (V - V_S)^2 \Theta(V - V_S). \quad (3.22)$$

Consider the vacuum coordinate above, for the von Neumann entropy in (3.4) and the backreacted dilaton in (3.14). We will be interested in the regime

$$\epsilon \ll |\alpha| \ll 1. \quad (3.23)$$

The extremization of the generalized entropy (1.3) leads to the shifted island location

$$V_i = \frac{U_r \epsilon}{3} + 16\pi G_2 \alpha \frac{V_S \epsilon}{c} + \mathcal{O}(\epsilon^2, \alpha^2 \epsilon), \quad (3.24)$$

$$U_i = \frac{V_r \epsilon}{3} - 16\pi G_2 \alpha \frac{\epsilon}{c V_S} + \mathcal{O}(\epsilon^2, \alpha^2 \epsilon), \quad (3.25)$$

for which, the von Neumann entropy collected on $\mathbf{R}$ due to the island is given as

$$\begin{aligned} S_{\text{isl}} = & \frac{\phi_0}{G_2} + \frac{c}{6} \left(1 - \frac{3}{\epsilon} + 2 \log \frac{4\ell^4}{\epsilon_i \epsilon_r (\ell^2 - U_r V_r)} \right) \\ & + 16\pi G_2 \alpha + \frac{\epsilon (c V_S V_r U_r + 48\pi G_2 \alpha (V_S^2 V_r - \ell^2 U_r))}{9\ell^2 V_S}. \end{aligned} \quad (3.26)$$

The result agrees with [17, 72] when $\alpha = 0$.

We now consider the situation illustrated in figure 5 where $V_r U_r \approx \ell^2$ (corresponding to $\mathbf{R}$ being close to $\mathcal{I}^\pm$). (3.26) to $\mathcal{O}(\alpha)$ in the regime (3.23) becomes

$$S_{\text{isl}} \simeq S_{\text{BH}} + \frac{c}{3} \log \frac{4}{\epsilon_i \epsilon_r \delta L^2} + 16\pi G_2 \alpha, \quad (3.27)$$

where $\delta L^2 = \ell^2 - U_r V_r$, and $S_{\text{BH}} = \frac{1}{2G_2}(\phi_0 - \phi_r)$ is the Bekenstein-Hawking entropy of the Nariai black hole. Notice that (3.27) provides a bound on the amount of information that can be transmitted between the observers through the island. The classical regime corresponds to $S_{\text{isl}} \simeq S_{\text{BH}}$, similar to (2.22). However, notice that we are not comparing information bounds due to the opening of the traversable wormhole in the SdS₄ geometry in section 2 with the von Neumann entropy derived from the island rule in the setup shown in figure 5. The former relies on a classical bulk geometry in 4-dimensional gravity, while the latter relies on a semiclassical theory whose matter content is inherently 2-dimensional, despite the fact that the geometry comes from the s-wave reduction of a higher dimensional near-Nariai black hole spacetime.

As we repeat the procedure, the total null shift of the light rays intercepted by the north pole observer in figure 5 would then correspond to (3.18) with $\alpha \rightarrow \alpha' < 0$ for the net null shift, and the condition of intercepting the island corresponds to $U_r < U_{\alpha'}$.

4 Discussion and outlook

In summary, we proposed a new protocol to form traversable wormholes in SdS black holes. Due to the lack of a holographic description, a conceptual difficulty in the protocol arises. Consider generating a traversable wormhole without adding a coupling between antipodal static patch observers in the near-Nariai black hole. It might seem that one can produce a signal between causally disconnected regions when there is no proper synchronization of the observers to emit the shockwaves responsible for the traversable wormhole. For this reason, we started from the Bunch-Davis state of dS space as the initial configuration. The antipodal observers are then maximally entangled, and by employing a classical communication channel, their clocks can then be synchronized. We also require that the pair of observers store energy coming from the Hawking radiation of the cosmological horizon, so they can later collapse the energy to form a near-Nariai black hole. Later, we proposed a symmetric protocol, where each of the observers throws an excess of positive energy into the cosmological horizon, which results in a negative energy excess into the black hole horizon. Then, a traversable wormhole will form in the black hole interior, as well as in the inflationary region. Each observer can send light signals at the appropriate time, that can cross through the newly formed wormhole. Thus, they are able to apply an appropriate recovery channel and decode the information by an appropriate protocol. In order to quantify such recovery, we analyzed bounds on the amount of information that can be transmitted through the wormhole. In the second part of the manuscript, we incorporated semiclassical effects explicitly by establishing the protocol in dS JT gravity and probing the island region. We considered how information can be transmitted through a wormhole in the island according to the metaobserver in the inflating region. Notice that our protocol does not advocate a particular notion of dS holography, and it might as well provide new hints towards a yet-to-be-known holographic dual theory to the SdS space-time. In addition, when not restricted to a shockwave description ($E/M \ll 1$), the maximal amount of bits that can be transferred from one static patch to the other is equal to the total number of bits contained in *empty* dS space. Naively, one could have guessed that solely the entropy of the black hole would provide a bound on the possible information exchange ($\Delta S_b = S_N = \text{Area}(r_N)/(4G_N)$). However, releasing M_N of energy

will not only destroy the Nariai black hole but it will also extend the cosmological horizon as well (since $\Delta S_c = S_{dS} - \text{Area}(r_N)/(4G_N)$).

We now comment on some extensions of our work.

Other extensions with the island formula. One of the main drives in this work was incorporating bulk matter in the background to probe the island region with our protocol. This was explicitly performed in $(1+1)$ -dimensions, but we expect it can be similarly done in higher dimensions. To study such effects in an explicit setting, perhaps double holographic models with dS braneworlds [75, 76] would be a fruitful future direction. One might be able to incorporate quantum SdS black holes [77, 78] in such a setting to treat quantum backreaction effects exactly.

Moreover, the 2-dimensional analysis was based on the s-wave reduction of a near-Nariai black hole configuration with shockwave sources, to simplify the evaluation. However, given the instability of this configuration, one should in principle describe the evaporation of arbitrary dS black holes from dS JT gravity. Modeling the evaporation of dS black holes in dS JT gravity has been made in [54, 56]. There has been a lot of progress in information recovery during an adiabatic evaporation regime with free particle bulk matter fields [79] that could be extended to SdS space.

An alternative analysis of the islands could be pursued with the set-up of [57]. In that model, the radiation is collected in a subregion of asymptotically flat space coupled to the dS JT gravity model, instead of considering the detectors near $\mathcal{I}^+$ of the inflating patches, as it was studied here. It might be possible to reproduce our protocol in this setting with better analytic control over backreaction effects. In [57], one can encounter the islands in the inflating patch of dS₂ space due to backreaction effects, which do not affect the observers in the asymptotically flat region. Perhaps this setting would allow us to study a traversable wormhole on the island in the inflating region which we have discussed in section 3.

States and algebra of observables. In the more general case of a SdS black hole, there is a conceptual difficulty in properly defining states given that the cosmological horizon and the black hole horizon are not in thermal equilibrium. In this case, the construction of Hilbert spaces can enter into ambiguities for defining thermofield double states and studying the algebra of semiclassical states on the SdS background. It would be very interesting to make a rigorous treatment of quantum states of these types of black holes, based on recent developments in von Neumann algebras [66, 80, 81] (see also [82] where an algebraic approach to a teleportation protocol in AdS black holes has been proposed). For our protocol, it might be possible to make a TFD-type state preparation of modes coming from the near-Nariai black hole and cosmological horizons, as well as the modes contained in the thermal baths. We leave a more rigorous treatment of the quantum states involved in our protocol for future directions. Moreover, it would be interesting to study whether the preparation of a vacuum state can be extended in the presence of conical defects for general SdS black holes.

Stretched horizon holography. Let us reiterate that our protocol does not employ a particular realization of dS space holography. However, stretched horizon holography [83] might allow for different modifications. In this framework, the stretched horizon represents a

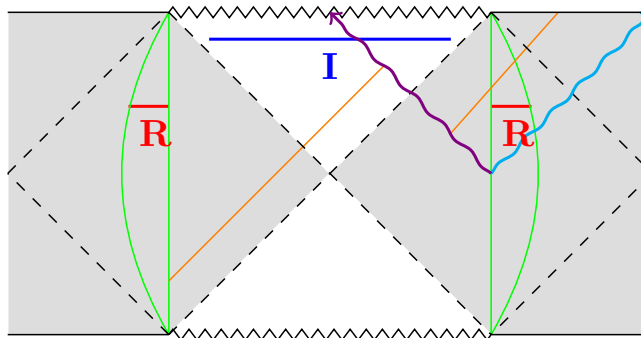


Figure 6. Alternative realization of the protocol employing the stretched horizon holographic proposal of [88].

spacetime surface very close to the cosmological horizon where the holographic dual to dS space is located. It has been conjectured that the doubled-scaled SYK model might display some features of dS JT gravity [84–86] at infinite temperatures, and it might be located at its stretched horizon. This approach mirrors several aspects of the AdS/CFT correspondence in that observables can be gravitationally dressed with respect to this surface.

It might be possible to perform the protocol in this framework following the same steps as illustrated in the protocol above according to the initial state preparation in empty dS space¹² and transmitting information in SdS black hole backgrounds. For probing the island region, we might consider an alternative configuration with respect to figure 5, which we illustrate in figure 6. In this case, the pair of detectors are no longer located at $\mathcal{I}^+$, but rather on the stretched horizon, extending to the static patch observers. We expect the qualitative results would be just as the found for the configuration in figure 5 for recovering information about the island region. It would also be interesting to study if a realization of the protocol could be performed explicitly in a double-scaled SYK model in the limit that might reproduce dS JT gravity [84]. There have been different realizations of the GJW protocol in different contexts of the SYK [10, 11, 89, 90] that might be useful to pursue this direction.

Traversable wormhole production rates. Recently, [91] provided the false vacuum decay rate of a pair of Reissner-Nordstrom (RN) AdS_4 black holes into traversable wormholes connecting the geometries. In the microcanonical ensemble, they find that

$$\Gamma \sim e^{\Delta S + 2\beta(M_b - M_0)}, \quad (4.1)$$

where ΔS is the total entropy of the pair of RN AdS_4 black holes, β refers to their (inverse) temperature, M_b the black hole mass, and M_0 the mass of the traversable wormhole. It would be very interesting to study this type of process in the context of our protocol. Given that we required opposite energy pulses to produce the wormhole, we expect the contribution of the traversable wormhole from the black hole and the cosmological horizon patches will approximately cancel each other in the near-Nariai configuration, such that (4.1) might

¹²Notice that the worldline observers in pure dS space, figure 1, would also have access to an island near the cosmological horizon [87], which has been noticed to have different puzzling features, such as it moving backward in time [50] and violating entanglement wedge nesting [87].

be modified to

$$\Gamma \sim e^{\Delta S + 2\beta_N M_N}, \quad (4.2)$$

according to the static patch observers. We expect this result since thermal fluctuations can also open traversable wormholes between the black hole and inflating regions for Nariai spacetimes. However, the size of such wormholes would not be significant enough to be of practical use. It might be useful to study the quasi-local thermodynamics approach to cosmological and black hole horizons [72, 92–94] to make this relation precise.

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A Hayden-Preskill protocol for near-Nariai black holes

We refer the reader to [22] for an introduction to Hayden-Preskill protocol and information recovery in pure dS space, as well as the original publication [95] in the black hole context.

The basic principle of operation of this protocol is illustrated in figure 7. From purely quantum information arguments, we search for a criterion establishing if the information transfer between the antipodal observers in the near-Nariai black hole spacetime can be successful.

We consider figure 7 where one of the observers, Alice, possesses a subsystem $\mathcal{M}$ which is maximally entangled with a subsystem $\mathcal{N}$ carried on by Charlie. Meanwhile, an observer Bob has access to subsystem $\mathcal{E}$ (representing Hawking radiation) which is entangled with a black hole, $\mathcal{B}$. Alice waits until $\mathcal{E}$ is maximally entangled with $\mathcal{B}$ when she throws $\mathcal{M}$ into $\mathcal{B}$. We model the absorption of the message as a random unitary process V^B . This results in new subsystems, $\mathcal{B}'$ describing the black hole after absorption, and $\mathcal{R}$ for the radiation. In that process, $\mathcal{M}$, the purifier of $\mathcal{N}$, gets transferred from the subsystem $\mathcal{B}'$

$$|\psi\rangle = \frac{1}{\sqrt{\|\mathcal{B}'\|}} \sum_i |i\rangle_{\mathcal{B}'} |0\rangle_{\mathcal{R}} |i\rangle_{\mathcal{N}}, \quad (A.1)$$

to the subsystem $\mathcal{R}$

$$|\psi'\rangle = \frac{1}{\sqrt{\|\mathcal{B}'\|}} \sum_i |0\rangle_{\mathcal{B}'} |i\rangle_{\mathcal{R}} |i\rangle_{\mathcal{N}}, \quad (A.2)$$

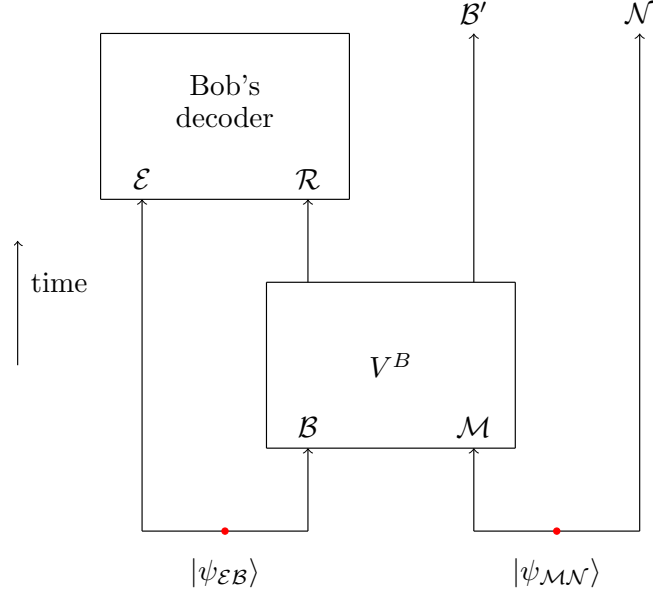


Figure 7. Quantum circuit representation of the Hayden-Preskill protocol, where $|\psi_{EB}\rangle$ and $|\psi_{MN}\rangle$ are maximally entangled states. Figure based on [95].

where $\|B'\|$ refers to the Hilbert space dimension of B' . The criterion in [95] for successful message transmission is that

$$\|\rho_{MB'} - \rho_M \otimes \rho_{B'}\| \ll 1, \quad (\text{A.3})$$

where ρ_M , $\rho_{B'}$, $\rho_{MB'}$ refer to the reduced density matrices by tracing out the subsystems $B'\mathcal{N}$, $\mathcal{N}M$ and $\mathcal{N}$. The norm is taken with respect to the Haar measure, defined as

$$\int dV^B \|\rho_{MB'} - \rho_M \otimes \rho_{B'}\| = 2^{-2C}, \quad (\text{A.4})$$

with C denoting the additional qubits in the radiation with respect to the number of qubits in Alice's message. This means that as soon as the black hole has radiated more information than the amount encoded in the Alice's original message, Bob gains access to it.

We can directly apply this protocol to transmit light signals in the near-Nariai black hole background in figure 2. However, notice that in this case, Alice does not need to wait until the Page time to throw $\mathcal{M}$ into $\mathcal{B}$. The reason is that the preparation of the state from the BD vacuum already maximally entangles the pair of antipodal observers. As we saw in section 2, we achieved information recovery by transmitting signals across the traversable wormhole produced in the near-Nariai black hole over a time scale determined by (2.17). Meanwhile, if we perform the information transfer in the inflationary region, the time scale for recovery is given by (2.20).

The reader is referred to [22, 95] for considerations about how to avoid the cloning paradox in HP protocol for the black holes and in pure dS space.

B S-wave reduction of SdS black holes

We follow the derivation of the full-reduction dS JT gravity model given [72]. In our notation, hats are used for $(d+1)$ -dimensional quantities and unhatted for $(1+1)$ -dimensional ones.

Starting from Einstein gravity in $(d + 1)$ -dimensions

$$I = \frac{1}{16\pi G_N} \int_{\mathcal{M}} d^{d+1}X \sqrt{-\hat{g}}(\hat{R} - 2\Lambda) + \frac{1}{8\pi G_N} \int_{\partial\mathcal{M}} d^dY \sqrt{-\hat{h}} \hat{K}, \quad (\text{B.1})$$

with a positive cosmological constant

$$\Lambda = \frac{d(d-1)}{2\ell^2} > 0. \quad (\text{B.2})$$

The general solution is given by the SdS metric (2.2). We consider dimensional reduction with the metric ansatz,

$$ds^2 = g_{\mu\nu}(x)dx^\mu dx^\nu + r_N^2 \Phi^{\frac{2}{d-1}} d\Omega_{d-1}^2 \quad (\text{B.3})$$

where $\mu, \nu = 0, 1$, and $\Phi(x)$ is the dilaton. Next, we perform a Weyl rescaling $g_{\mu\nu} \rightarrow \frac{1}{\sqrt{d}} \Phi^{-\frac{d-2}{2(d-1)}} g_{\mu\nu}$ to express the EFT coming from the spherical reduction of SdS black holes as:

$$I = \frac{1}{16\pi G_2} \int_{\mathcal{M}} d^2x \sqrt{-g} [\Phi R + U(\Phi)] + \frac{1}{8\pi G_2} \int_{\partial\mathcal{M}} \sqrt{-h} \Phi K, \quad (\text{B.4})$$

with

$$U(\Phi) = \frac{d-1}{\ell^2} \left(\Phi^{-\frac{1}{d-1}} - \Phi^{\frac{1}{d-1}} \right), \quad (\text{B.5})$$

$$G_2 = \frac{G_N}{\ell^{d-1} \Omega_{d-1}}.$$

The resulting equations of motion are

$$(\nabla_\mu \nabla_\nu - g_{\mu\nu} \nabla^2) \Phi + \frac{g_{\mu\nu}}{2} U(\Phi) = -8\pi G_2 T_{\mu\nu}^{(m)}, \quad (\text{B.6})$$

$$R = U'(\Phi), \quad (\text{B.7})$$

where we added a $(1 + 1)$ -dimensional bulk-matter term I_m generating the matter stress-tensor $T_{\mu\nu}^{(m)}$. We also notice that the minimum of the potential (B.5) is located at $\Phi = 1$, corresponding to the Nariai black hole limit, for which

$$U(\Phi = 1) = 0, \quad (\text{B.8})$$

$$\left. \frac{dU}{d\Phi} \right|_{\Phi=1} = -\frac{2}{\ell^2}.$$

If we take $\Phi = \phi_0 + \phi$, with $\phi_0 = 1$, we can perform a series expansion for $\phi \ll 1$ and truncate the action (B.4) to first order in ϕ we recover the so-called dS₂ JT full-reduction model with the minimally coupled bulk matter fields that we introduced in (B.6)

$$I_{\text{dSJT}} = I_0 + \frac{1}{16\pi G_2} \int_{\mathcal{M}} d^2x \sqrt{-g} \left(\phi R - \frac{2}{\ell^2} \phi \right) + \frac{1}{8\pi G_2} \int_{\partial\mathcal{M}} dy \sqrt{-h} \phi K + I_{\text{matter}}, \quad (\text{B.9})$$

where $I_0 = \frac{\phi_0}{16\pi G_2} \int_{\mathcal{M}} d^2x \sqrt{-g} R$ refers to the topological term in the action, and I_{matter} is the matter content. Notice that the addition of the dilaton mode breaks asymptotic dS₂ space isometries, namely reparametrizations of coordinates at $\mathcal{I}^+$ [96].

C Shifts from metric gluing

In a seminal work by Shenker and Stanford [97], a convenient way to obtain the shift in trajectory experienced by a probe message following a null-like geodesic was stated. We extend this formalism to SdS space. Let us express the Kruskal coordinates in (3.6) in terms of static patch coordinates (2.2), (2.3):

$$\begin{aligned} U_{b,c} &= e^{\frac{f'(r_{b,c})}{2}(r_*(r)-t)}, \\ V_{b,c} &= -e^{\frac{f'(r_{b,c})}{2}(r_*(r)+t)}, \end{aligned} \quad (C.1)$$

centered at the horizons $r_{b,c}$ for $0 \leq r < r_{\mathcal{O}}$ and $r_{\mathcal{O}} \leq r < \infty$ respectively. Evaluating the tortoise coordinate $dr_* = dr/f(r)$ results in

$$ds^2 = \begin{cases} -\frac{4f(r)}{f'(r_b)} e^{-f'(r_b)r_*(r)} dU_b dV_b + r^2 d\Omega_{d-1}^2 \equiv ds_b^2 & 0 \leq r < r_{\mathcal{O}} \\ -\frac{4f(r)}{f'(r_c)} e^{-f'(r_c)r_*(r)} dU_c dV_c + r^2 d\Omega_{d-1}^2 \equiv ds_c^2 & r_{\mathcal{O}} \leq r < \infty \end{cases}. \quad (C.2)$$

When gluing two SdS space-times of different masses to each other, both the black hole and cosmological line elements will experience shifts along $U = U_0$ and $V = V_0$ respectively,

$$\begin{aligned} ds_b^2 &= \begin{cases} ds_b^2 & U \leq U_0 \\ d\tilde{s}_b^2 & U_0 < U \end{cases}, \\ ds_c^2 &= \begin{cases} ds_c^2 & V \leq V_0 \\ d\tilde{s}_c^2 & V_0 < V \end{cases}, \end{aligned} \quad (C.3)$$

where we simplified the notation to $(U_{b,c}, V_{b,c}) \rightarrow (U, V)$ for more clarity. We focus on the black hole region $0 \leq r < r_{\mathcal{O}}$, and introduce a perturbation at $t = t_0$ forming a shell propagating along the surface

$$\begin{aligned} \tilde{U}_0 &= e^{\frac{\tilde{f}'(\tilde{r}_b)}{2}(\tilde{r}_*(r_{\mathcal{O}})-t_0)}, \\ U_0 &= e^{\frac{f'(r_b)}{2}(r_*(r_{\mathcal{O}})-t_0)}, \end{aligned} \quad (C.4)$$

such that the gluing/matching condition relating $\tilde{V}$ to V can be expressed as follows

$$\begin{aligned} \tilde{U}_0 \tilde{V} &= -e^{\tilde{f}'(\tilde{r}_b)\tilde{r}_*(r)}, \\ U_0 V &= -e^{f'(r_b)r_*(r)}. \end{aligned} \quad (C.5)$$

When working in the small energy limit $E/M = \epsilon \ll 1$, $\tilde{U}_0$ can be well approximated by U_0 , allowing us to deduce that, at first order, the shift along the V coordinate; which can be expressed

$$\begin{aligned} \tilde{V} &= V - \frac{\epsilon}{U_0} \left(\frac{\partial}{\partial \epsilon} e^{\tilde{f}'(\tilde{r}_b)\tilde{r}_*(r)} \right)_{\epsilon=0} \\ &\equiv V + \alpha. \end{aligned} \quad (C.6)$$

As was pointed out in [97], $e^{f'(r_b)r_*(r)}$ can be compactly written (in $d = 3$) as

$$\begin{aligned} e^{f'(r_b)r_*(r)} &= \left(\frac{r-r_b}{\ell}\right) \left(\frac{r-r_c}{\ell}\right) - \frac{r_c(2r_b+r_c)}{r_b(r_b+2r_c)} \left(\frac{r_b+r_c+r}{\ell}\right) \frac{r_c^2-r_b^2}{r_b(r_b+2r_c)} \\ &\equiv (r-r_b) C_b(r, r_b), \end{aligned} \quad (C.7)$$

and the shift α becomes

$$\alpha = \frac{G_N E \beta_b}{2\pi r_b U_0} \frac{\partial}{\partial r_b} \left((r_b - r) C_b(r, r_b) \right), \quad (C.8)$$

where we used $\tilde{M} = M(1 - \epsilon)$ and we normalized the temperature with respect of the observer radius.¹³ Obviously, the same is happening in the cosmological region $r_{\mathcal{O}} \leq r < \infty$ (along V). Thus, the coordinate shifts can be expressed as

$$\begin{aligned} \alpha_b &= \frac{r_b}{2U_0} \frac{s_b}{S_b} \frac{\partial}{\partial r_b} \left((r_b - r) C_b(r, r_b) \right), \\ \alpha_c &= \frac{r_c}{2V_0} \frac{s_c}{S_c} \frac{\partial}{\partial r_c} \left((r_c - r) C_c(r, r_c) \right), \end{aligned} \quad (C.10)$$

where we restored the indices and introduced the wormhole entropies $s_{b,c} = E\beta_{b,c}$. Notably, in the Nariai limit $r_{b,c} \rightarrow r_N$ the function $C_N(r, r_N) = (r - r_N)^{-1}$. This means that the shifts tend to vanish. However, considering $M \neq M_N$ and inserting the shockwave far in the past, the value for r is getting pushed towards the respective horizon radius. We then find

$$\begin{aligned} \alpha_b &= \frac{r_b}{2U_0} \frac{s_b}{S_b} C_b(r_b, r_b), \\ \alpha_c &= \frac{r_c}{2V_0} \frac{s_c}{S_c} C_c(r_c, r_c). \end{aligned} \quad (C.11)$$

Notice, again, that a probe message sent towards the black hole will experience a different shift than a probe message sent towards the cosmological horizon when encountering shockwaves of the same magnitude of energy, E .

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¹³Using the normalized temperature in $dM = TdS$ allows to write

$$\frac{\partial r_h}{\partial M} = \frac{\partial r_h}{\partial S_h} \frac{\partial S_h}{\partial M} = \frac{\beta_h G_N}{2\pi r_b}. \quad (C.9)$$

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